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Vertical-axis block rotations in southwestern China since the Cretaceous: new paleomagnetic results from Hainan Island

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Abstract. A paleomagnetic study of Early Cretaceous redbeds from Hainan Island, South China, reveals a pre-folding remanence from six sites, with a mean direction of D=4.3°, I=41.8°, k=51.5, α_{95}=9.4°. The results suggest that Hainan Island may have been rotated ~10° counterclockwise about a vertical axis since the Early Cretaceous. This rotation is likely to have occurred during the mid-Tertiary, when large scale left-lateral motion occurred along the Red River Fault as a result of the indentation of the Indian Plate into Eurasia causing extrusion of the Indochina Block, and opening of the South China Sea. A comparison of Early Cretaceous paleomagnetic results from South China indicates that local rotations are widespread in the western part of the South China Block. Such rotations have to be taken into account when examining pre-Cenozoic tectonic kinematics of South China, and constructing apparent polar wander path(s) for the region.

Introduction

In analogue modelling of the India–Asia collision [e.g., *Tapponnier et al.*, 1982], the South China Block was normally taken, to a first order, as a coherent crustal block. Paleomagnetic studies, however, suggest that local Tertiary rotations occurred in both regions close to the cratonic margin [e.g., *Enkin et al.*, 1991b], and in some interior parts of the craton [e.g., *Gilder et al.*, 1993].

Vertical-axis rotations may have been widespread in the South China Block due to either the pervasive mid-Mesozoic orogenies, or the Indian Plate's northward indentation in the Cenozoic. To better understand the tectonic kinematics of East Asia since the Mesozoic, it is critical to determine firstly the Tertiary tectonic rotations within the South China Block through paleomagnetic studies. Cretaceous strata in South China are ideally suited for such studies because of their age of deposition (after the mid-Mesozoic orogeny, but just prior to India–Asia collision), their lithology (mostly redbeds), and their gentle tectonic deformation.

Field Sampling and Laboratory Techniques

Samples for this study came from Hainan Island, located off the southern Chinese coast (Figure 1). The island has a metamorphic basement of 1.4–1.0 Ga, and a variably metamorphosed Neoproterozoic succession [*Zhang et al.*, 1990]. Paleozoic successions were mostly cleaved, but the Early Cretaceous terrestrial conglomerate and redbeds (the Lumuwan Formation), which are over 2600 m thick and well exposed in generally NE trending basins (Figure 1), are only gently deformed. Ostracoda and charophyte fossils indicate a late-Early Cretaceous age for the upper Lumuwan Formation [*Zhang et al*, 1992], from which our samples were collected.

Sixty-one core samples, oriented using both magnetic and sun compasses, were collected. All samples were processed at the paleomagnetic laboratory of the University of Western Australia. MMTD1 thermal demagnetizers manufactured by MAGNETIC MEASUREMENTS were used for stepwise thermal demagnetisation. A 2G 755-R superconducting rock magnetometer with an attached 2G600 automatic AF degauss system were used for AF demagnetisation and remanence measurement. All instruments are hosted in a low-field space controlled by three sets of 3.8m x 3.8m Helmholtz coils. The principal component analysis technique of Kirschvink [1980] was used for analysing the demagnetisation data.

Results and Interpretation

In sites 1–4, 6 and 7, characteristic remanent magnetisations were revealed from all but one sample after the removal of a small viscous component. Hematite is regarded as the prime remanence carrier, as indicated by high coercivity (Figure 2a) and

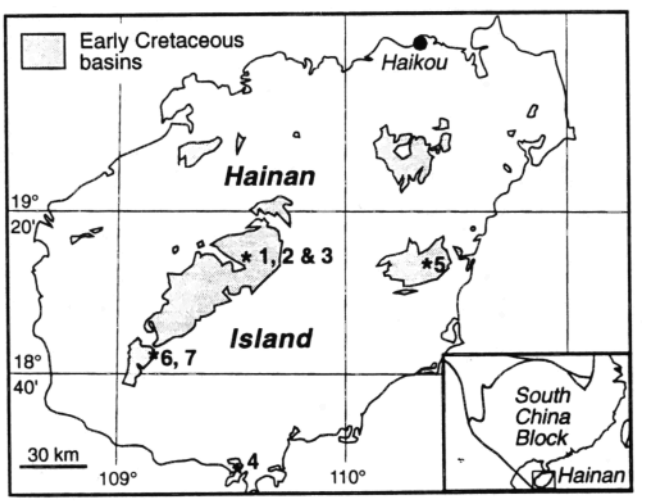


Figure 1. Geographic location of the study area. Numbers indicate sampling localities.

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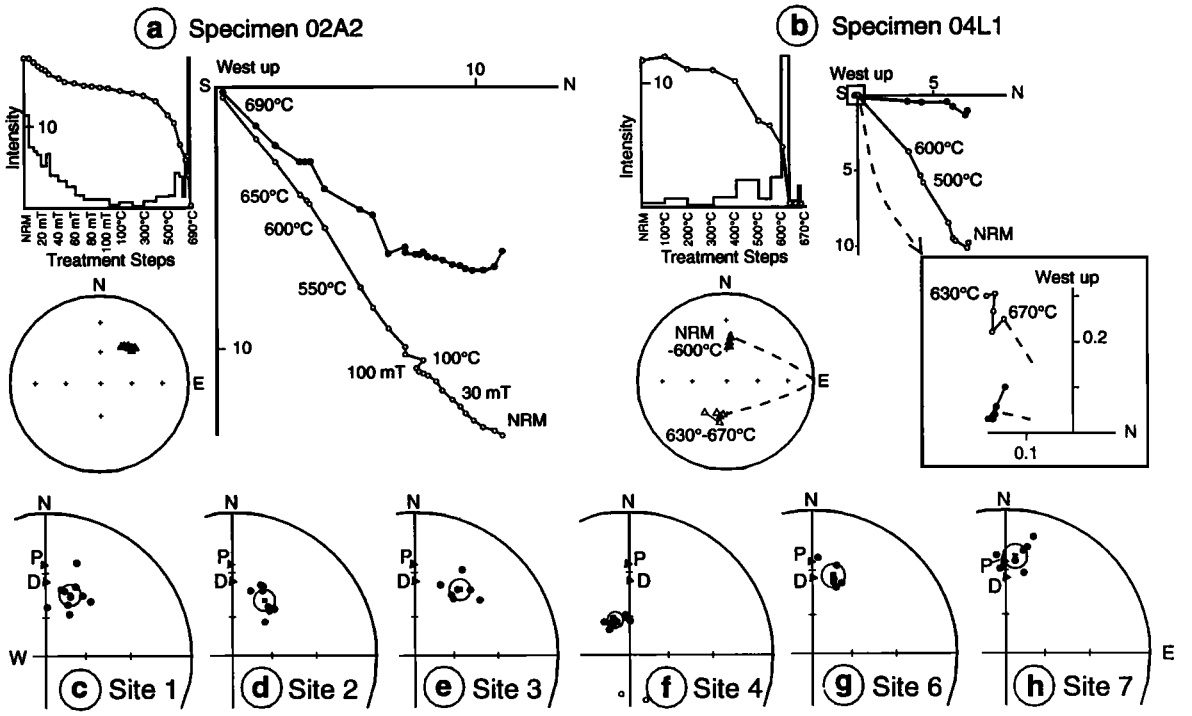


Figure 2. Representative plots of demagnetisation results. Remanence intensity units are in mA/m. P—present local geomagnetic field direction; D—present dipole field direction. Filled (open) symbols are projections on the lower (upper) hemispheres.

the >600°C unblocking temperature (Figure 2a & b). The *in-situ* directions within each of the six sites are well grouped, with most of their site-means clearly departing from the present geomagnetic field direction (G) and dipole field direction (D) (Figure 2 c–g). In site 4, four of the 15 samples indicate the existence of dual polarities, and in two of the samples the reversed component can be clearly identified (Figure 2b & f). Site 5 was from a severely weathered dam-bottom. It revealed a characteristic magnetic remanence with D = 356.3°, I = 44.0°, n = 6, α_{95} =11.2°, which is sub-parallel to the present dipole field direction before bedding-correction, and departs from the rest of the site-means after bedding-correction. This site is regarded as having a recent weathering-induced overprint only, and is excluded from further data analysis.

Figure 3 demonstrates that the site-mean directions group better after structural correction. A McFadden [1990] fold test (definition 1) gives a 95% critical value SCOS = 2.86, in-situ SCOS1 = 3.88, and bedding-corrected SCOS1=2.26, again indicating a possible pre-folding origin. Although the maximum k (= 71.7) is achieved at 73% unfolding, the predominant normal polarity of the remanence further suggests it was likely acquired during the mid-Cretaceous normal supercron. The results are listed in Table 1.

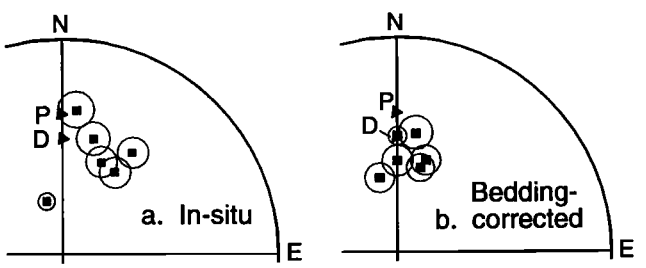


Figure 3. Stereographic projections of the site-mean directions (a) before and (b) after bedding-correction

Counterclockwise Rotation of Hainan Island, and Opening of the South China Sea

Figure 4 presents published Early Cretaceous paleomagnetic results from South China. It is apparent that results from the

Table 1. Site-mean directions and paleopole positions of the interpreted primary remanent magnetisation

Site No.	Locality (°N)	(°E)	Bedding (Dip/Azim.)	n (N)	In-situ mean Dec (°)	Inc (°)	Bedding-corrected Dec (°)	Inc (°)	Common k	α_{95}	Pole position (°N)	(°E)	dp	dm
1	19.1	109.5	34.5/257.9	10	21.7	41.0	345.6	51.1	47.8	7.1	71.9	66.7	6.5	9.6
2	19.1	109.5	32.5/281.0	8	30.7	42.0	359.0	44.3	64.1	7.0	83.0	102.1	5.5	8.8
3	19.3	109.5	28.0/262.0	8	33.5	31.0	14.0	46.2	78.5	6.3	74.8	164.2	5.2	8.1
4	18.3	109.5	31.0/021.0	15	341.5	60.5	359.0	33.4	92.2	4.0	89.0	16.5	2.6	4.6
6	18.8	109.2	10.0/284.7	6	15.0	31.8	8.8	31.3	94.9	6.9	81.4	210.5	4.3	7.7
7	18.8	109.2	26.7/149.0	8	5.6	22.0	17.0	42.0	69.3	6.7	73.2	177.3	5.0	8.2
		Mean		(6)	16.8	39.0	4.3	41.8			83.2	143.0		
					k = 18.9	α_{95} = 15.8	k = 51.5	α_{95} = 9.4			K= 47.7	A_{95} =9.8		

n (N) = number of samples (sites).

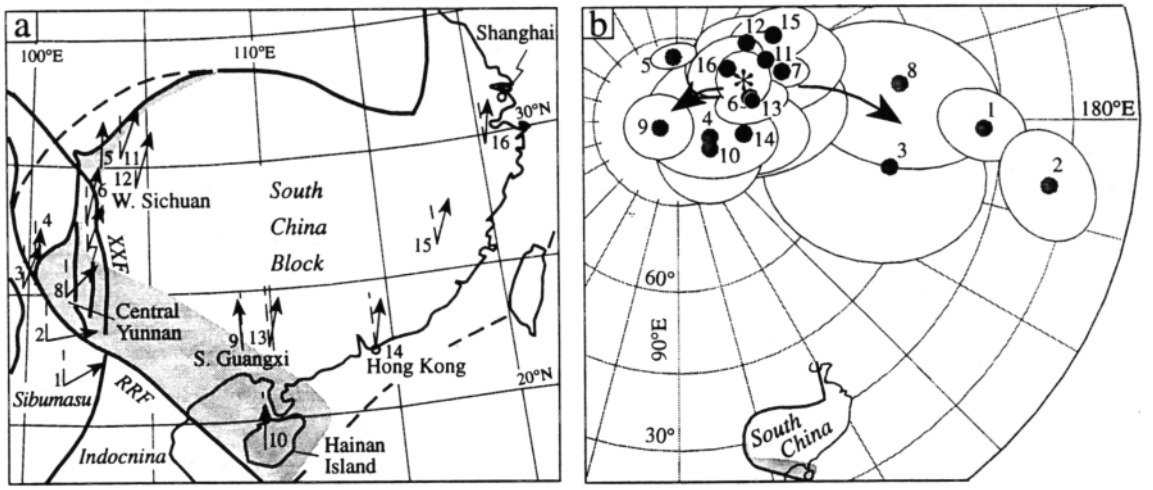


Figure 4. Plots of Early Cretaceous paleomagnetic results from South China demonstrating local/regional rotations in western and southwestern South China (shaded region). Paleomagnetic poles from the rotated region are shown as shaded dots in (b). XXF = Xianshuihe-Xiaojiang fault zone; RRF = Red River Fault. Star in (b) represents the reference pole for Eurasia (excluding Chinese data) [*Enkin et al.*, 1992]. Sources of data are: 1, 2 & 4, Huang and Opdyke [1993]; 3, Funahara et al. [1993]; 5, Enkin et al., 1991b; 6, Zhu et al. [1988]; 7, Huang and Opdyke [1992]; 8, Funahara et al. [1992]; 9 & 13, Gilder et al. [1993]; 10, this study; 11 & 12, Enkin et al. [1991a]; 14, Chan [1991]; 15, Zhai et al. [1992]; 16, Lin et al. [1985].

eastern part of South China are in general agreement with each other, and their corresponding paleopoles agree with that of the Eurasia reference pole [*Enkin et al.*, 1992]. However, results from western South China are mostly discordant. Apart from one locality on the Red River Fault [No. 4; *Huang and Opdyke*, 1993], results from both sides of the fault in western Yunnan Province indicate clockwise rotations relative to the eastern South China Block and Eurasia. Clockwise block-rotations in the region west of the Red River Fault are consistent with predictions of the indentation model [e.g., *Tapponnier et al.*, 1982], although the amount of rotation in the Sibumasu (Shan-Thai) Block, western Yunnan (46–65°) [*Huang and Opdyke*, 1993] is significantly larger than that of the Indochina Block (14° ± 7°) [*Yang and Besse*, 1993].

Results from both Hainan Island [10, this study] and from southern Guangxi Province [9, *Gilder et al.*, 1993] indicate a small (10–20°), but consistent counterclockwise rotation relative to the eastern part of the South China Block. Gilder et al. [1993] interpreted the counterclockwise rotation in southern Guangxi Province as a regional block rotation, rather than rotation within a fault zone. This also appears to be the case in Hainan Island, as our samples were from both the central part and south end of the island (Figure 1), and no Cenozoic transcurrent fault zone has been reported on the island. Such a rotation is consistent with the model prediction that the extrusion of Indochina relative to South China during the late Eocene–mid-Miocene [*Schärer et al.*, 1994] had led to the opening of the South China Sea [e.g., *Briais et al.*, 1993]. The left-lateral movement along the Red River Fault caused the rifted crustal blocks to the east of it to rotate counterclockwise [*Briais et al.*, 1993] (Figure 5a). Conversely, paleomagnetic data do not support the alternative slab-pulling model for the opening of the South China Sea as suggested by Holloway [1982] and Taylor and Hayes [1983], which would instead produce clockwise rotations in the region (Figure 5b).

Conclusions

Early Cretaceous paleomagnetic results indicate a 10–20° Cenozoic counterclockwise block rotation of Hainan Island and surrounding regions relative to central and eastern South China.

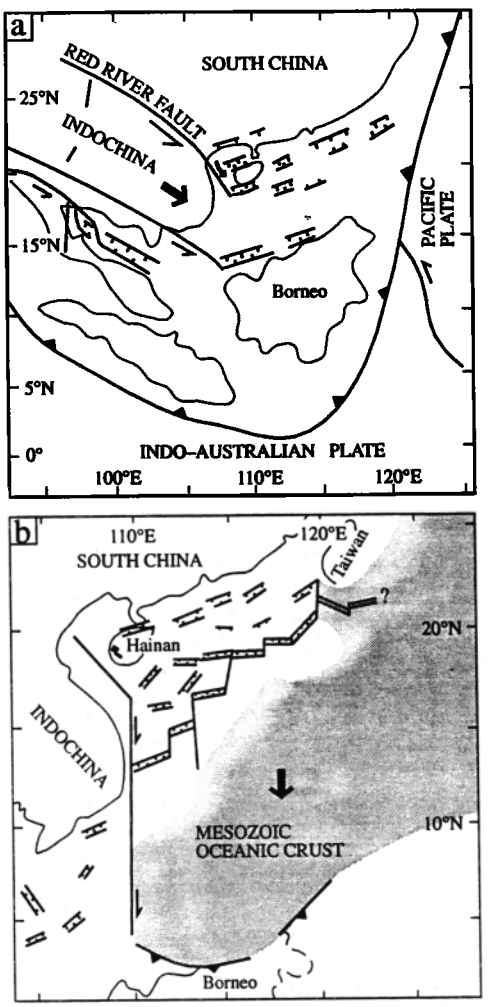


Figure 5. Schematic sketch of tectonic models proposed for the opening of the South China Sea in the mid-Eocene to mid-Oligocene. Model (a) [after *Briais et al.*, 1993] predicts counterclockwise rotations in Hainan region, whereas model (b) [after *Taylor and Hayes*, 1983] predicts clockwise rotation [*Briais et al.*, 1993].

This rotation is interpreted as a result of the mid-Tertiary left-lateral movement along the Red River Fault during the extrusion of Indochina, which also has been interpreted to have caused the opening of the South China Sea [*Briais et al.*, 1993].

Paleomagnetic results suggest that post-Cretaceous vertical-axis rotations are widespread in the western and southwestern parts of the South China Block. These rotations not only occur within fault zones (such as the Xianshuihe–Xiaojiang fault system, the Longmenshan thrust system and the Red River fault system), but are also recorded in crustal blocks of hundreds of kilometres in dimension (e.g., the central Yunnan region and the southern Guangxi–Hainan Island region). Pre-Cretaceous paleomagnetic results from these regions need to be corrected for such rotations before being applied to the rest of the South China Block.

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